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Modeling fiscal-monetary regime switching in Poland

Bachelor's Thesis
in Economics

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Oświadczenia

Oświadczenie kierującego pracą

Oświadczam, że niniejsza praca została przygotowana pod moim kierunkiem i stwierdzam, że spełnia ona warunki do przedstawienia jej w postępowaniu o nadanie tytułu zawodowego.

Data

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Świadom odpowiedzialności prawnej oświadczam, że niniejsza praca dyplomowa została napisana przeze mnie samodzielnie i nie zawiera treści uzyskanych w sposób niezgodny z obowiązującymi przepisami.

Oświadczam również, że przedstawiona praca nie była wcześniej przedmiotem procedur związanych z uzyskaniem tytułu zawodowego w wyższej uczelni.

Oświadczam ponadto, że niniejsza wersja pracy jest identyczna z załączoną wersją elektroniczną.

Data

Podpis autora pracy

Abstract:

This thesis investigates the existence of monetary- and fiscally-led policy regimes in Poland between 2000 and 2025 using Markov-switching macroeconomic models. First, a Markov-Switching VAR model is estimated to identify possible regime changes in inflation, output growth, interest rates, debt, and the primary balance. Second, a Markov-Switching DSGE model with rational expectations is estimated within the RISE framework.

The results suggest the presence of two distinct regimes. The first is characterized by stronger monetary responses to inflation and greater fiscal discipline, while the second exhibits weaker monetary reactions and more persistent fiscal imbalances. The estimated probabilities indicate that fiscally-led dynamics became more likely during periods of economic stress, particularly in 2019–2022. Impulse response analysis further shows that the transmission of monetary and fiscal shocks differs substantially across regimes.

Keywords:

fiscal dominance, monetary dominance, Markov-switching VAR, Markov-switching DSGE

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Title in Polish:

Modelowanie zmian reżimów fiskalno-monetarnych w Polsce.

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ARTYKUŁ NAUKOWY

”Sustained high inflation is always
and everywhere a fiscal phenomenon,
in which the central bank is a
monetary accomplice”

– Tom Sargent

1 Introduction

The recent inflationary episode—the most significant in the past four decades—has attracted substantial attention in the economic literature. As Reis (2026) argues, several conventional explanations have been proposed. Following the end of pandemic-related restrictions, workers returned to the labor market, while governments provided additional support to household incomes through extensive fiscal transfers. At the same time, global supply chains remained severely disrupted due to the breakdown of international trade routes. The interaction of these demand- and supply-side shocks generated strong inflationary pressures, which were further intensified after Russia’s full-scale invasion of Ukraine in 2022, as emphasized by Powell (2025) in his Jackson Hole speech.

The role of fiscal policy—and the reluctance of monetary authorities to raise interest rates in the aftermath of the combined demand and supply shocks, partly out of concern that tighter monetary conditions could undermine post-Covid relief and recovery programs—has also become the subject of extensive debate. The issue of rising public debt has long occupied a central place in macroeconomic discussions; however, recent years have renewed interest in the concepts of fiscal and monetary dominance, the possibility of regime changes between them, and the macroeconomic consequences associated with such shifts.

This paper builds on the assumption, shared by many economists (Leeper 1991, Sims 1994, Woodford 1994), that government debt and its management play a central role in shaping real economic outcomes. The existence of fiscal and monetary regimes, together with their historical decomposition, has already been documented for the United States (Bianchi & Melosi 2017) and the United Kingdom (Webb 2025). However, no comparable analysis has yet been conducted for Poland. Although several studies have examined regime switching within Taylor-rule specifications, none has combined monetary–fiscal interactions with the possibility of multiple equilibria in a DSGE framework. Consequently, this paper constitutes the first attempt to identify and quantify the existence of such regimes in the Polish economy.

The main objective of this study is to determine whether regime changes occurred in Poland, to quantify the probability of their existence, and to evaluate how different regimes affected key macroeconomic variables, namely interest rates, inflation, real GDP, and debt. The

next section reviews the existing literature on policy regimes, their classification, and quantitative analyses of their macroeconomic implications. Subsequently, two models are developed. First, an MS-VAR model is estimated in order to obtain priors for the Markov chain governing regime transitions and to examine whether the Polish economy exhibits regime-switching behavior. Second, the main empirical framework of the paper, an MS-DSGE model, is introduced. The subsequent section presents the results, including calibrated parameters, impulse response functions across regimes, which helps to explain the main differences between models. The final section offers conclusion and additional thoughts.

2 Literature

The discussion on the importance of fiscal policy for the effectiveness of monetary policy can be traced back to Sargent & Wallace (1981). Even under the standard monetarist assumption of exogenous real interest rates—implying that monetary authorities cannot permanently maintain unemployment below its natural level—the control of inflation is not guaranteed. If the government accumulates debt financed through persistent fiscal deficits, the central bank may eventually be forced to accommodate these liabilities through money creation, thereby losing effective control over the inflation rate.

This contribution laid the foundations for Leeper (1991), who organized possible policy regimes into four categories: (i) active monetary policy and active fiscal policy, (ii) active monetary policy and passive fiscal policy, (iii) passive monetary policy and active fiscal policy, and (iv) passive monetary policy and passive fiscal policy. Regimes (i) and (iv) are either indeterminate or unstable and can therefore be excluded from further analysis. By contrast, regime (ii), commonly referred to as the monetary-led (ML) regime, and regime (iii), known as the fiscally-led (FL) regime, are associated with unique equilibria. Leeper derived the theoretical properties of these equilibria and established the conditions under which the two regimes can be distinguished.

Research on the existence and probability of such regimes has generally followed two distinct approaches. The first, employed by Sims (2010), involves estimating a Markov-Switching Vector Autoregression (MS-VAR) model, in which particular attention is paid to the relationships between deficits and debt, as well as between interest rates and inflation. This methodology was subsequently applied by Kliem (2016) to the United States, Germany, and Italy, and by Bianchi & Melosi (2017) to the United States. These studies were able to identify clear differences in central bank responses to inflation. A central focus of their analyses, however, concerned the threat posed by the Zero Lower Bound (ZLB), namely a situation in which nominal interest rates approach zero while inflation and output remain persistently below target levels, as occurred in the United States following the Global Financial Crisis. During this period, the government pursued aggressive fiscal stimulus while the Federal Reserve temporarily tolerated deviations from its standard policy objectives. Nevertheless, Bianchi & Melosi (2019) warn about the macroeconomic costs associated with conflicts between monetary and fiscal authorities, episodes that have occurred several times throughout U.S. history and were typically accompanied by lower growth and higher inflation.

The literature on fiscal and monetary rules for Poland, including regime-switching behavior, also provides an important foundation for the calibration and interpretation of the model developed in this paper. Brzoza-Brzezina, Kolasa & Szetela (2016) estimated a DSGE model for Poland, and several of their calibrated parameters are employed in the present study. Their work focused primarily on the possibility of Poland reaching the Zero Lower Bound, an event

that would likely correspond to a period of fiscal dominance. However, they concluded that such a scenario was relatively unlikely for Poland, given the country's comparatively strong growth performance and a higher inflation target than in many Western European economies, which reduced the risk of persistent deflation. Their analysis, however, did not incorporate time-varying coefficients or regime-switching behavior. Additional parameters and priors used in this paper are based on Grabek, Klos, Koloch (2011). The issue of switching regimes was also examined by Baranowski & Sztaudynger (2016), who developed a Markov-switching Taylor rule for Poland. Their analysis did not address fiscal dominance directly, but they found evidence that the central bank engaged in changes in the responsiveness of monetary policy to inflation and the output gap over time. Nevertheless, the differences between regimes were relatively moderate, and the authors did not identify an underlying transition mechanism beyond the estimated evolution of policy coefficients.

The methods used to estimate MS-DSGE models are also well established in the literature. For standard DSGE models, Herbst & Schorfheide (2016) provide a comprehensive introduction to the construction of Dynamic Stochastic General Equilibrium models, Bayesian inference, and Markov Chain Monte Carlo techniques, with particular emphasis on the Metropolis–Hastings algorithm for DSGE estimation. These methods are employed in both the MS-VAR and MS-DSGE frameworks developed in this paper. With respect to the interaction between fiscal and monetary policy in DSGE models with switching regimes, Sims, Zha (2006) provide an important methodological contribution using U.S. data. Their framework was further refined by Bianchi & Melosi (2017), whose Bayesian estimation strategy—combining calibrated parameters with prior distributions—serves as the principal methodological foundation for the present study.

The current mainstream approach in DSGE modelling relies on rational expectations (RE), both in forward-looking and hybrid specifications. However, standard rational expectations frameworks are not fully suitable for models with regime switching, since economic agents also account for the possibility of future regime changes when forming expectations. To address this issue, Maih (2015) introduced Rationality in Switching Environments (RISE), a framework that allows rational expectations to be formed even when model parameters vary across regimes. The key idea is that agents recognize that regimes evolve according to a Markov process and therefore incorporate the probability of regime transitions into their expectations and decision-making processes.

3 Data

The empirical analysis is based on five quarterly macroeconomic time series: the inflation rate, the real GDP growth rate, the short-term nominal interest rate, the ratio of government debt to GDP, and the ratio of the primary surplus to GDP. The sample spans the period from 2000-Q2 to 2025-Q4, yielding a total of 103 quarterly observations. The starting point of the sample is determined by the construction of the inflation series, which requires the previous quarter's price level and therefore cannot be computed for 2000-Q1.

The short-term nominal interest rate is represented by the three-month interbank rate for Poland (FRED series: IR3TIB01PLQ156N). Government indebtedness is measured using quarterly general government debt data obtained from Eurostat (Eurostat series: gov_10q_ggdebt). It should be noted that this series is affected by several structural breaks resulting from methodological and institutional changes, particularly major fiscal reforms. The most important example is the 2013 pension reform, under which a substantial share of OFE pension fund assets and liabilities was transferred outside the general government sector, leading to a discrete decline in the reported debt level.

Inflation and real output growth are constructed as log differences of the GDP deflator (FRED series: NGDPDSAIXPLQ) and real GDP (FRED series: NGDPRSAXDCPLQ), respectively. Since the model assumes the existence of a non-zero steady-state growth path, the cyclical component of real GDP growth is extracted using the Hodrick–Prescott filter in order to isolate deviations from trend growth.

The primary surplus series is constructed from general government net lending/borrowing data (FRED series: GGNLBAPLA188N). Because this measure includes government interest payments, net interest expenditures are removed in order to obtain the primary balance. Net interest payments are calculated as the difference between government interest expenditures and interest revenues using Eurostat data (Eurostat series: gov_10q_ggnfa). The resulting series is subsequently seasonally adjusted to eliminate the pronounced seasonal component present in fiscal variables.

All observable variables are demeaned or transformed by removing a steady state (this is the case for inflation, for which I assume the quarterly steady state at 0.625%). The variables used, along with their description, are described in the table 1.

Table 1: Variables used in the analysis

Variable	Description	Source	Mean
Inflation	Log difference of GDP deflator	FRED	0.78%
Real GDP growth	HP-filtered log difference of real GDP	FRED	0.9%
Short-term interest rate	90-day interbank rate	FRED	1.24%
Debt-to-GDP ratio	Quarterly general government debt	Eurostat	49.42%
Primary surplus-to-GDP ratio	General government net lending/borrowing adjusted for net interest payments	FRED / Eurostat	-0.53%

4 Model

The following section is divided into 2 parts: firstly a MS-VAR in the spirit of Bianchi & Melosi (2017) is created and calibrated for Polish data, which serves as a guide to create the full MS-DSGE model, which is presented in the second part.

4.1 Markov-Switching VAR

In order to check my intuition regarding the existence of regime switching in Poland, which is realized with distinct reaction of monetary authority to inflation and fiscal authority to debt I use four observables to estimate the Markov-Switching Vector Autoregression (MS-VAR): (i) inflation; (ii) real GDP growth rate; (iii) real overnight interest rate ex ante; (iv) ratio of primary surplus to current debt.

I allow for Markov-switching regimes for the parameters of VAR matrix and for the variance-covariance matrix (Σ_t). I estimate the model looking for two regimes Φ_{ξ_t} : each with its own Φ_t and associated σ_t . My main purpose is finding the VAR parameters (especially the persistence of surplus/debt [for fiscal block] and relationship between interest rate and inflation [for monetary block]). The estimated model can be characterized by the set of equations:

$$Z_t = c_{\xi_t} + A_{\xi_t} Z_{t-1} + \Sigma_{\xi_t}^{\frac{1}{2}} \varepsilon_t \quad (1)$$

$$\Phi_{\xi_t} = (c_{\xi_t}, A_{\xi_t}, \Sigma_{\xi_t}), \varepsilon_t \sim N(0, I) \quad (2)$$

4.2 New Keynesian Model

In this section I introduce a Markov Switching Dynamic Stochastic General Equilibrium model, which is augmented typical New Keynesian (NK) model with added fiscal block and the Markov process that generates two regimes of monetary-fiscal policy. This NK model, based on the model used by Bianchi & Melosi (2017), involves representative agents: households, firms and government, which has 2 branches: monetary that controls the (nominal) interest rate and fiscal which controls spending.

I have also decided to use fully Rational Expectations in my model in line with Galí (2008), however with a change regarding switching regimes. The used RISE model, as presented by Maih (2015) assumes that agents (whose expectations matter) are aware of the existing Markov process ruling the regimes and form their expectations accordingly.

4.2.1 Households

A representative household maximizes expected lifetime utility:

$$\max_{\{C_t, N_t, B_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right) \quad (3)$$

subject to the nominal budget constraint:

$$P_t C_t + B_t \leq W_t N_t + (1 + i_{t-1}) B_{t-1} + \Pi_t - T_t \quad (4)$$

and the market clearing of the whole economy:

$$Y_t = C_t + G_t \quad (5)$$

We also assume that the product (and all variables, as their steady states are dependent on the product) follows a deterministic trend with stochastic variations around the steady state, namely:

$$Y_t = (1 + dy)^t \bar{Y} (1 + \tilde{y}_t) \quad (6)$$

Following the log-linearization around the steady state of growth, we get the IS-curve:

$$\tilde{y}_t = E_t(\tilde{y}_{t+1}) - \frac{1}{\sigma}(\hat{i}_t - E_t(\hat{\pi}_{t+1})) + \gamma_g(\tilde{g}_t - E_t\tilde{g}_{t+1}) + \zeta_t \quad (7)$$

where ζ_t is an supply shock process, $\hat{i}_t$, $\hat{\pi}_{t+1}$ are (linearized) deviations from steady state nominal interest rate and from inflation target, respectively, while $\hat{g}_t$ is log-linearized deviation from the steady state level of government spending.

4.2.2 Firms

Firms declare their demand for work (which clears the labour market) and with this create product:

$$Y_t = (1 + dy)^t N_t \quad (8)$$

Price setting follows the Calvo (1983) mechanism. Each period, a fraction of firms receives permission to re-optimize their price; the remaining fraction keep their current price unchanged. A firm choosing P_t^* solves:

$$\max_{P_t^*} E_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} [P_t^* Y_{t+k|t} - P_{t+k} MC_{t+k} Y_{t+k|t}] \quad (9)$$

The resulting New Keynesian Philips Curve yields (after log-linearization):

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \kappa \tilde{y}_t + u_t \quad (10)$$

where:

$$\kappa = \frac{(1 - \theta)(1 - \beta\theta)}{\theta} (1 + \varphi) \quad (11)$$

and u_t is a cost-push shock process, coming from random shocks and falling with certain persistence ρ_u .

4.2.3 Fiscal-monetary block

The monetary authority reacts to inflation, as well as an output gap. Both of those reactions are dependent on the regime in place - in the monetary regime the reaction to both is larger than in the fiscal regime, because the monetary authority is passive then. The Taylor rule can therefore be written as:

$$i_t = \pi^* + r^* + \phi_\pi^{\xi_t} (\pi_t - \pi^*) + \phi_y^{\xi_t} (y_t - dy) + u_t \quad (12)$$

,

which around the steady state (and with certain persistence (ρ_i) to central bank decisions, which is consistent with real decisions) can be expressed as:

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + (1 - \rho_i) (\phi_\pi^{\xi_t} \hat{\pi}_t + \phi_y^{\xi_t} \tilde{y}_t) + \varepsilon_t^i \quad (13)$$

where ε_t^i represents the monetary policy shocks - the changes in interest rate not consistent with the rule.

In turn government collects taxes every period, issues new debt, repays the debt outstanding and spends the money as a government consumption.

$$B_t = (1 + i_{t-1})B_{t-1} + P_t G_t - P_t T_t \quad (14)$$

which may be presented in real terms and in relationship to product:

$$b_t = (1 + i_{t-1} - \pi_t - dy_t)b_{t-1} + g_t - t_t \quad (15)$$

When dealing with situation around the steady state (where $g_t = t_t$) I assume for the sake of simplicity that government keeps the taxes steady and allows only for the movement in spending. I also assumed that the government reacts countercyclically - raises deficit and government spending in crisis and decreases it in a times of high growth. As such, the rule for surplus (which is also a rule for government expenditures) is as follows:

$$-s_t = \tilde{g}_t = \rho_g \tilde{g}_{t-1} - (1 - \rho_g) [\delta_g^{\xi_t} (b_t - b^*) - \delta_g \tilde{y}_t] + \varepsilon_t^g \quad (16)$$

The used ξ_t represents the current regime.

4.2.4 Final model

The resulting New Keynesian model may be altogether given in a set of 7 equations:

- New Keynesian IS curve

$$\tilde{y}_t = E_t(\tilde{y}_{t+1}) - \frac{1}{\sigma}(\hat{i}_t - E_t(\hat{\pi}_{t+1})) + \gamma'_g(\tilde{g}_t - E_t\tilde{g}_{t+1}) + \zeta_t \quad (17)$$

- New Keynesian Philips curve

$$\hat{\pi}_t = \beta(1 + dy)^{\frac{1}{\sigma}} E_t\hat{\pi}_{t+1} + \kappa\tilde{y}_t + u_t \quad (18)$$

- Monetary rule

$$\hat{i}_t = \rho_i\hat{i}_{t+1} + (1 - \rho_i)(\phi_{\pi}^{\xi_t}\hat{\pi}_t + \phi_y^{\xi_t}\tilde{y}_t) + \varepsilon_t^{MP} \quad (19)$$

- Fiscal rule

$$\tilde{g}_t = \rho_g\tilde{g}_{t-1} - \delta_b^{\xi_t}\hat{b}_t - \delta_g\tilde{y}_t + \varepsilon_t^g \quad (20)$$

- Evolution of government debt

$$\hat{b}_t = (1 + i_{ss} - \pi_{ss} - dy)\hat{b}_{t-1} + (i_{ss} + \hat{i}_{t-1} - \pi_{ss} - \hat{\pi}_t - dy - \tilde{y}_t + \tilde{y}_{t-1})b_{ss} + g_{ss} + \tilde{g}_t \quad (21)$$

- Evolution of demand shock

$$\zeta_t = \rho_{\zeta}\zeta_{t-1} + \varepsilon_t^{\zeta} \quad (22)$$

- Evolution of cost-push shock

$$u_t = \rho_u u_{t-1} + \varepsilon_t^u \quad (23)$$

My target has been to calibrate some parameters (namely: risk aversion (σ), household discount factor (β), Philips curve kink (κ)) from literature, set some from the policy rules (steady-state inflation (π^*), response of government expenditure to debt in fiscal regime ($\phi_{\pi}^{\xi_t=1}$) and average quarterly GDP growth (dy)) from the previous estimations, while at the same time estimating the rest of them from the available data.

Two additional shocks has been added to help the calibration and make estimation better: the Covid-19 shock (which was set from Q2-2020 to Q2-2021) was added, with negative impact on product and positive impact on inflation. The size of this shock was estimated in the model. This way, proposed by Smets (2026) was used in order to reduce the effects of Covid-19 pandemic on the estimation. Additionally, the debt shock was also used to calibrate the size

of government debt, in order to account for the structural breaks, of which I speak in the Data section.

5 Results

5.1 Calibration of MS-VAR

In order to solve MS-VAR analytically, I created Bayesian VAR with Minnesota prior and then iterated $N=1M$ times in order to get posterior results. Assuming the existence of the 2 regimes, I was able to find the parameters for both and estimate the probability of each one being in place.

The resulting smoothed probabilities of the regimes in MS-VAR following burnout of 10% of posterior results are visible in the 1. The lighter variation corresponds to the regime 1, while the darker to the regime 2.

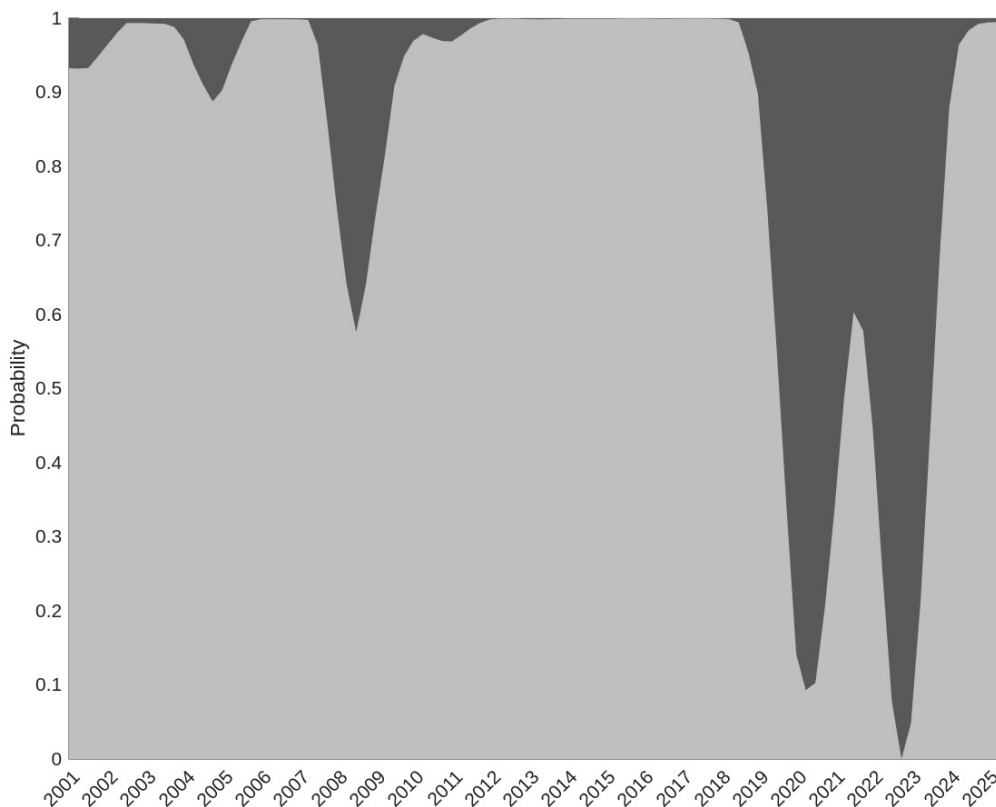


Figure 1: Posterior probability of policy regimes in MS-VAR

As for the regimes found: the results of the estimates for the most important parameters are given in the table 2. Regime 1 has a very strong response of real rate to inflation, which means aggressive Taylor rule coefficients. It is also connected with net surplus constant, as well as higher persistence of this variable.

Table 2: Estimated Regime-Specific VAR Coefficients

Coefficient	Regime 1	Regime 2
Real rate response to inflation $\phi_{r,\pi}$	0.9521	0.1012
Surplus/deficit persistence $\phi_{f,f}$	0.9903	0.1801
Surplus/deficit constant $\phi_{f,0}$	0.0226	-0.3881

Additional information can be found by looking at macroeconomic variables in both regimes. Regime 1 is characterized by moderate inflation averaging 2.3%, strong real GDP growth of 3.8%, and positive real interest rates of 1.9%. In contrast, Regime 2 exhibits substantially higher inflation of 6.7%, weaker growth of 2.1%, and strongly negative real interest rates of -3.0%. Fiscal conditions also deteriorate in Regime 2, as indicated by a more negative surplus/debt ratio.

Table 3: Average Macroeconomic Conditions by Estimated Policy Regime

Variable	Regime 1	Regime 2
Inflation (%)	2.30	6.70
Real GDP Growth (%)	3.83	2.07
Real Interest Rate (%)	1.91	-3.00
Surplus/debt (%)	-3.96	-5.97

Taking into account that regime 1 corresponds to stronger reaction to both debt (which means stronger fiscal adjustments), as well as stronger reaction of real rates to inflation, while regime 2 shows neither a strong fiscal consolidation nor strong real rate reaction to inflation, I can assume that regime 1 corresponds to the monetary led (ML) regime, while regime 2 is connected with fiscally led (FL) regime.

Additionally, it is useful to include the resulting posterior probability of regime change matrix, presented in 5. Those values have later been used to create a prior for MS-DSGE.

Table 4: Posterior Transition Matrix for MS-VAR

$$P = \begin{bmatrix} 0.9881 & 0.0119 \\ 0.1299 & 0.8701 \end{bmatrix}$$

Considering all this information, MS-VAR gives us a strong reason to look into the period 2019-2023, where the strongest and most probable period of regime 2 (which I can interpret

as a sort of fiscal dominance) took place. However, because of lack of additional constraints it is not possible to determine the existence of different regimes using only MS-VAR. In order to check our findings, we have to engage in creating a general equilibrium model, that would include the fiscal-monetary block.

5.2 Calibration of MS-DSGE

The entire model has been created in RISE framework, with the set of 7 equations, 2 regimes and 24 variables, as well as 7 endogenous variables (5 of which are observable) and 5 exogenous variables (one of which is observable). Chosen parameters has been calibrated using the results of Grabek, Klos, Koloch (2011), while the rest has been given prior distribution. The full set of used priors and calibrated parameters is given in appendix. Following this process, the Markov Chain Monte Carlo (specifically Metropolis-Hastings algorithm) with $N=4M$ draws was done in order to get posterior distribution of the parameters and estimate the placing of the shocks.

The resulting smoothed probabilities of the regimes in MS-DSGE following burnout of 10% of posterior results are visible in the 2.

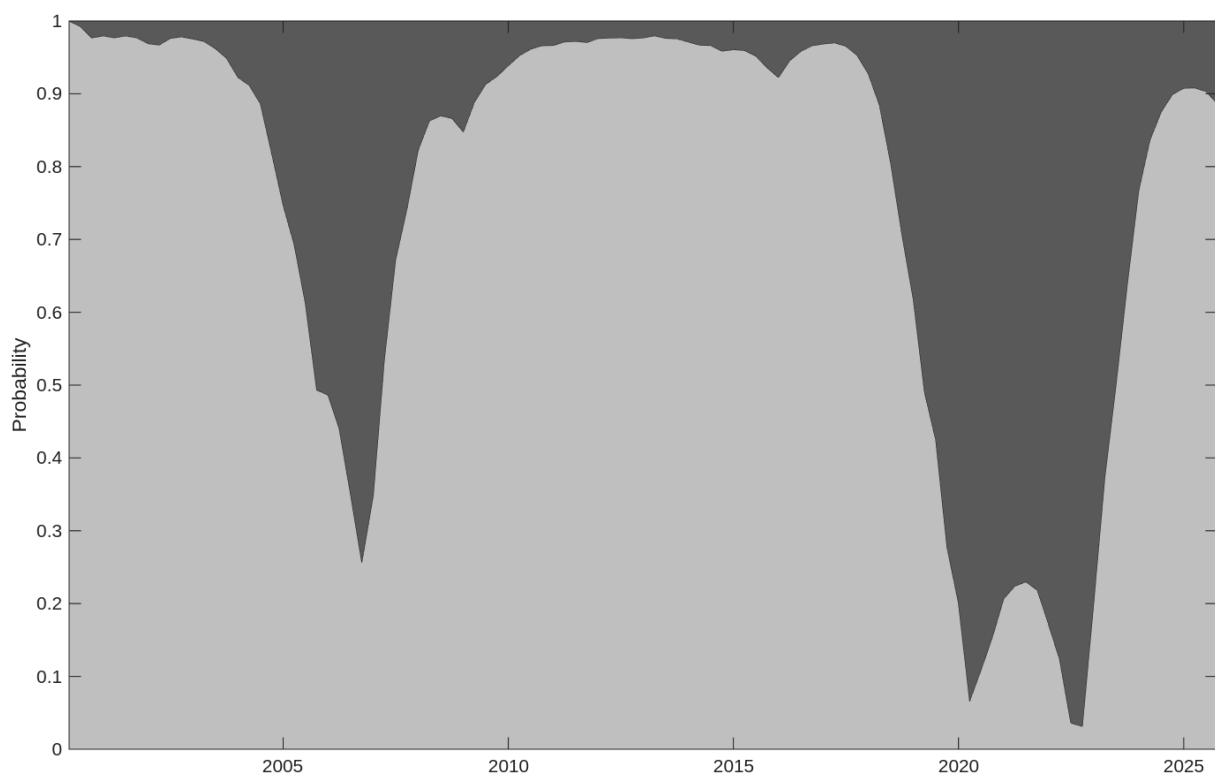


Figure 2: Posterior probability of policy regimes in MS-DSGE

Although the chance of either regime existing is never 0, the posterior probability of regime 2 (which is a monetary regime, because of the binding constraints on the coefficients) exceeds 50% twice. First time is around the year 2006 - following the entrance into European Union, the inflation rose, and the central bank deemed it transitory (Baranowski, Sztudynger

2016). More interesting is the second period - from Q12019 to Q42022, and reaches its maximum in Q22020. This is also to be expected - as Bianchi & Melosi (2019) have shown, the reaction to negative demand shock is often the switch to fiscal dominance.

It must also be noted, that both regimes are not symmetrical, when it comes to the probability. The estimated posterior transition matrix can be seen here:

Table 5: Posterior Transition Matrix for MS-DSGE

$$P = \begin{bmatrix} 0.9695 & 0.0305 \\ 0.1292 & 0.8708 \end{bmatrix}$$

As we can see, the dominant regime is regime 1 (monetary-led) - it is more stable and generally longer persistent than fiscally led regime, which may happen only during certain periods.

5.3 Impulse Response Functions

The classical way of showing influence of different regimes is presenting Impulse Response Functions - how different variables react to shocks of 1 s.d. In my model, I am especially interested in 4 shocks: monetary policy shock (ε_t^{MP}), fiscal policy shock (ε_t^g), demand shock (ε_t^ζ) and cost-push shock (ε_t^u).

The following picture shows the IRFs for the monetary regime (which I identify as regime 1):

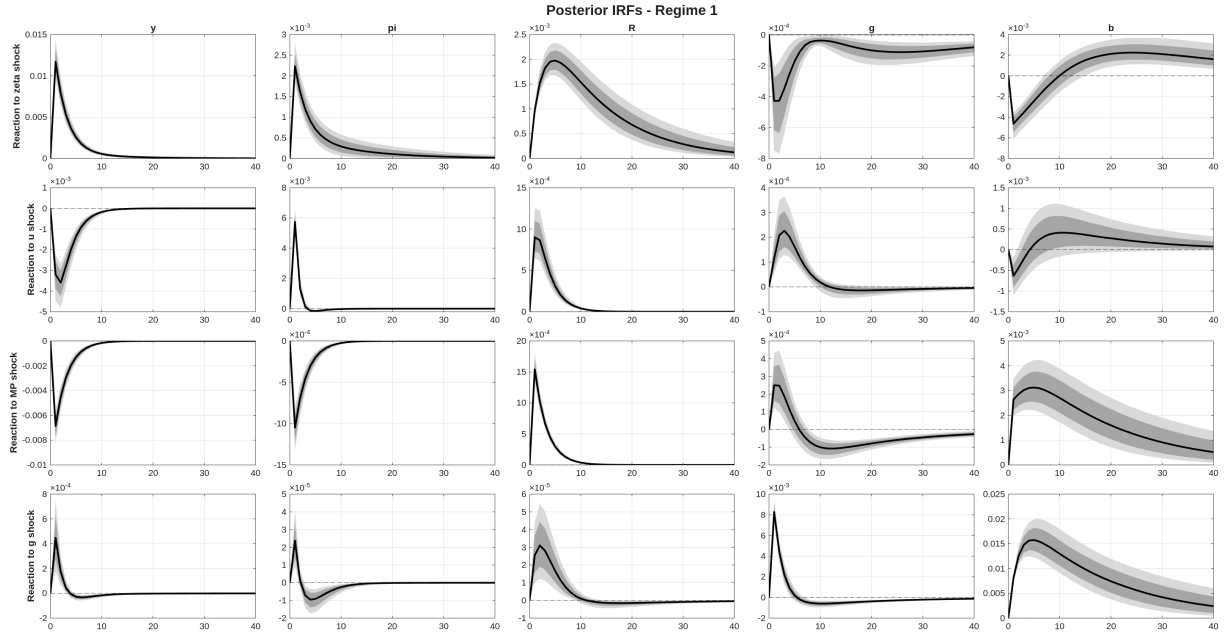


Figure 3: The Impulse Response Functions for 1 sd shocks to: demand shock, cosy-push shock, monetary policy shock and government spending shock in the monetary led regime

As is visible, the main reaction to happening crisis is, especially in the longer period, the interest rate. Although the fiscal reaction is also present, it is through the real rate channel that the economy is managed back into the steady state, while the government is busy keeping the debt in check. In the presence of inflation, the central bank rises interest rate more, thus keeping it in check.

This can be strongly contrasted with the IRFs for the fiscally regime (which can be identified as regime 2).

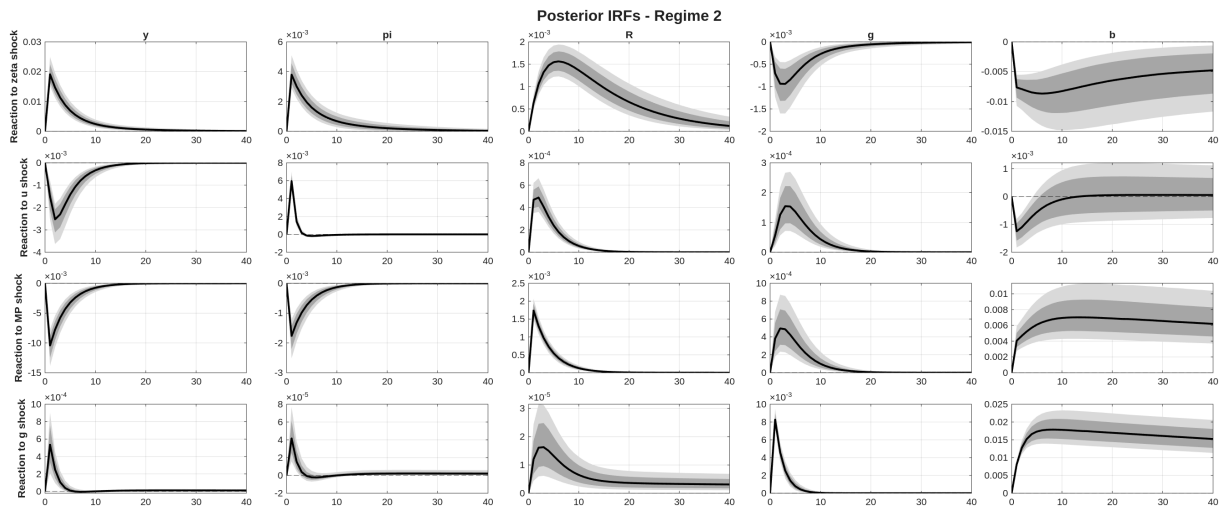


Figure 4: The Impulse Response Functions for 1 sd shocks to: demand shock, cosy-push shock, monetary policy shock and government spending shock in the fiscally led regime

In this regime, it is the fiscal policy that determines the price level. Inflation adjusts so that the government debt stays bounded. At the same time, the government does not try to balance the budget in order to keep the debt steady - it is inflation which does the job. Because of this, the interest rate grows less than inflation - as such, the growing inflation is a falling real rate. What is interesting is relatively high (around 0.8) posterior value of ϕ_π in fiscal regime. Poland never entered into institutional arrangement regarding the fiscal dominance (as was the case for the United States before 1951) - therefore, even in times of increased inflation the central bank reacts, though insufficiently.

5.4 Robustness checks

In order to check, whether the found solution was correct three parameters have been checked. Firstly, regarding the MCMC, the acceptance ratio was used, which theoretically should have been 0.234. In the data used, the parameter was about 0.26, which means that the Bayesian estimation was close to being theoretically correct. The analysis of prior and posterior distribution also shows significant differences, which suggest that the data is robust to the choice of priors.

Secondly, the results of the model and original data were plotted to compare them.

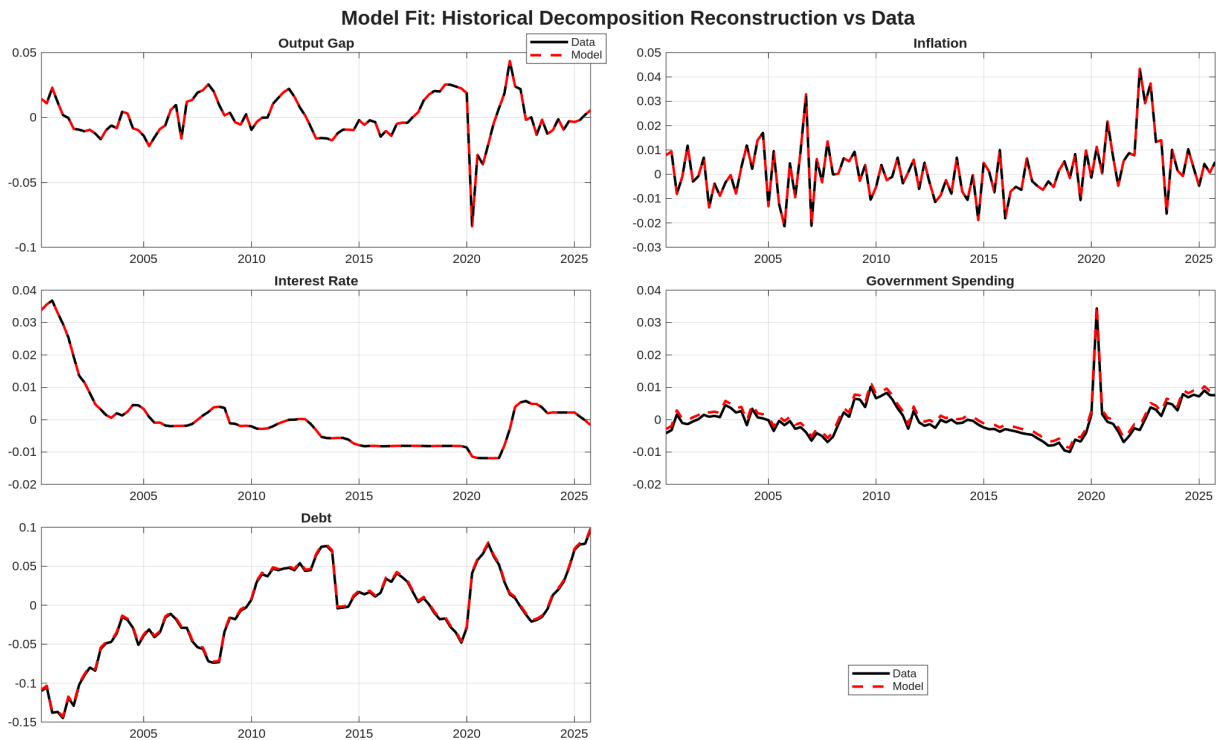


Figure 5: The estimated models compared to the data

As is visible, the model was able to replicate the collected data. The fit may be even too good - it may be a result of some form of overparametrization, which was rather negative. However, given the set of available data, there was no solution other than introducing non-structural shocks in order to increase the fit of model.

As the last robustness check serve the created IRFs. They are consistent with the literature - the reaction of interest rates, inflation and movement of debt across the regimes is in line with the literature regarding fiscal and monetary regimes. This serves to prove that results of the model are viable for analysis.

6 Conclusions

Both MS-VAR and MS-DSGE show a clear distinction between two possible regimes: the first was the standard in this period, while the second happened around the year 2006 as well as from 2019 to 2022. According to the model the differing regimes were, respectively, monetary led and fiscally led. During fiscally led periods, the inflation generally speaking rose, the central bank did not adjust the rates accordingly, while the general government allowed the high level of debt to endure.

The fiscal dominance is not always the same. Leeper (2023) himself distinguished 3 different parts of fiscally led regimes: unwitting dominance (in which the government want to keep the debt down, even when central banks try to activate economy, as following the “fiscal compact” from 2012 in the EU), insidious dominance (when the debt limit is changing, and yet government is not adjusting correctly, as is the case with rapidly aging countries) and political dominance (when the central bank is forced to act as a banker for the government to keep the bonds low, despite the rules saying otherwise). This analysis does not provide differentiation between those types of fiscal dominance periods in Poland.

It must also be noted, that this analysis is not centered on the welfare. Without it, it is not possible to decide whether regime change - which amplified inflation, forced the interest rate to remain at a higher level for longer and increased the government debt - was net positive or net negative. As Leeper (2023) himself notes: "There is nothing inherently “bad” about fiscal dominance or “good” about monetary dominance". It is very well possible, that switch from monetary led regime into fiscally led regime following Covid-19 pandemic or change in acceptable debt level following Great Recession has been a correct way to deal with threat of hitting zero lower bound (ZLB), as Bianchi & Melosi (2017) predict for the United States.

The main conclusion from this paper is that according to the model Poland entered into a period of fiscal dominance even before the Covid-19 pandemic and associated economic crisis, and was there throughout most of the crisis thereafter. The central bank did not respond strongly to inflation, while government disregarded the level of debt. This period ended following the strong monetary policy response to inflation in the course of 2022. However, this did not result in the fiscal consolidation, which may mean that the fiscal and monetary policies are currently in conflict, as is often the case following the crisis. As Bianchi & Melosi (2019) proved in their paper - the conflict between fiscal and monetary authority results in neither achieving their goal. Continued imbalances and increased lack of certainty regarding the ruling regime would be detrimental to macroeconomic stability.

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A VAR coefficients of MS-VAR

Table 6: Estimated Coefficient Matrix — Regime 1 (Monetary Dominance)

$$A_1 = \begin{bmatrix} -0.0008 & 0.8793 & 0.0254 & 0.0036 & -0.0002 \\ 0.0036 & -0.0130 & 0.7385 & -0.0135 & -0.0019 \\ 0.1976 & 0.9510 & 0.0013 & 1.0211 & 0.0141 \\ 0.0211 & -0.0032 & 0.0028 & -0.0023 & 0.9896 \end{bmatrix}$$

Table 7: Estimated Coefficient Matrix — Regime 2 (Fiscal Dominance)

$$A_2 = \begin{bmatrix} -0.0008 & 0.8793 & 0.0254 & 0.0036 & -0.0002 \\ 0.0036 & -0.0130 & 0.7385 & -0.0135 & -0.0019 \\ -0.2334 & 0.1005 & -0.0036 & 0.9794 & 0.0013 \\ -0.4379 & 0.0030 & -0.0038 & -0.0045 & 0.1668 \end{bmatrix}$$

B Log-linearization of New Keynesian model

Consider a representative household maximizing expected lifetime utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma} - 1}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right] \quad (24)$$

where:

- C_t denotes consumption,
- N_t denotes labor supply,
- $\sigma > 0$ is the inverse intertemporal elasticity of substitution,
- $\varphi > 0$ is the inverse Frisch elasticity,
- $\beta \in (0, 1)$ is the discount factor.

The household faces the nominal budget constraint:

$$P_t C_t + B_t = W_t N_t + (1 + i_{t-1}) B_{t-1} + \Pi_t - T_t \quad (25)$$

where:

- B_t denotes nominal government bonds,
- i_t is the nominal interest rate,
- W_t is the nominal wage,
- Π_t are profits,
- T_t are lump-sum taxes.

Define the stochastic discount factor:

$$\Lambda_t = C_t^{-\sigma} \quad (26)$$

The first-order condition for bond holdings is:

$$1 = \beta E_t \left[\frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} \frac{1 + i_t}{\Pi_{t+1}} \right] \quad (27)$$

where gross inflation is defined as:

$$\Pi_{t+1} = \frac{P_{t+1}}{P_t} \quad (28)$$

Hence:

$$1 = \beta E_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{1 + i_t}{\Pi_{t+1}} \right] \quad (29)$$

Assume deterministic technological growth:

$$A_t = (1 + dy)^t \quad (30)$$

Define stationary variables:

$$\tilde{C}_t = \frac{C_t}{A_t}, \quad \tilde{Y}_t = \frac{Y_t}{A_t} \quad (31)$$

Then:

$$\frac{C_{t+1}}{C_t} = (1 + dy) \frac{\tilde{C}_{t+1}}{\tilde{C}_t} \quad (32)$$

Substituting into the Euler equation gives:

$$1 = \beta (1 + dy)^{-\sigma} E_t \left[\left(\frac{\tilde{C}_{t+1}}{\tilde{C}_t} \right)^{-\sigma} \frac{1 + i_t}{\Pi_{t+1}} \right] \quad (33)$$

Define log deviations from steady state:

$$\tilde{y}_t = \log \left(\frac{\tilde{Y}_t}{\tilde{Y}} \right) \quad (34)$$

$$\hat{i}_t = i_t - i_{ss} \quad (35)$$

$$\hat{\pi}_t = \log(\Pi_t) - \pi_{ss} \quad (36)$$

Assuming market clearing:

$$\tilde{Y}_t = \tilde{C}_t + \gamma_g \tilde{G}_t \quad (37)$$

a first-order approximation yields the New Keynesian IS curve:

$$\tilde{y}_t = E_t(\tilde{y}_{t+1}) - \frac{1}{\sigma} (\hat{i}_t - E_t(\hat{\pi}_{t+1})) + \gamma'_g (\tilde{g}_t - E_t \tilde{g}_{t+1}) + \zeta_t \quad (38)$$

where ζ_t is a demand shock.

Final-good firms aggregate differentiated intermediate goods according to:

$$Y_t = \left(\int_0^1 Y_t(j)^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}} \quad (39)$$

Demand for intermediate good j is:

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\varepsilon} Y_t \quad (40)$$

Each period:

- a fraction $1 - \theta$ of firms can reset prices,
- a fraction θ keeps previous prices.

A firm resetting its price chooses P_t^* to maximize:

$$\max_{P_t^*} E_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} [P_t^* Y_{t+k|t} - P_{t+k} MC_{t+k} Y_{t+k|t}] \quad (41)$$

subject to:

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\varepsilon} Y_{t+k} \quad (42)$$

Log-linearization around the balanced-growth steady state yields:

$$\hat{\pi}_t = \beta(1 + dy)^{\frac{1}{\sigma}} E_t \hat{\pi}_{t+1} + \kappa \tilde{y}_t + u_t \quad (43)$$

where:

$$\kappa = \frac{(1 - \theta)(1 - \beta\theta)}{\theta} (\sigma + \varphi) \quad (44)$$

and u_t is a cost-push shock.

Assume a regime-switching Taylor rule:

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + (1 - \rho_i) (\phi_{\pi}^{\xi_t} \hat{\pi}_t + \phi_y^{\xi_t} \tilde{y}_t) + \varepsilon_t^{MP} \quad (45)$$

where:

$$\xi_t \in \{1, 2\} \quad (46)$$

denotes the monetary-fiscal regime.

Government spending evolves according to:

$$\tilde{g}_t = \rho_g \tilde{g}_{t-1} - \delta_b^{\xi_t} \hat{b}_t - \delta_g \tilde{y}_t + \varepsilon_t^g \quad (47)$$

where:

- $\delta_b^{\xi_t} > 0$ implies debt stabilization,
- $\delta_b^{\xi_t} \approx 0$ implies fiscal dominance.

The nominal government budget constraint is:

$$B_t = (1 + i_{t-1})B_{t-1} + P_t G_t - P_t S_t \quad (48)$$

Define debt-to-GDP ratio:

$$b_t = \frac{B_t}{P_t Y_t} \quad (49)$$

Then:

$$b_t = \frac{1 + i_{t-1}}{(1 + \pi_t)(1 + g_{y,t})} b_{t-1} + g_t - s_t \quad (50)$$

Linearizing around steady state yields:

$$\hat{b}_t = (1 + i_{ss} - \pi_{ss} - dy) \hat{b}_{t-1} \quad (51)$$

$$+ (i_{ss} + \hat{i}_{t-1} - \pi_{ss} - \hat{\pi}_t - dy - \tilde{y}_t + \tilde{y}_{t-1}) b_{ss} + g_{ss} + \tilde{g}_t \quad (52)$$

$$\zeta_t = \rho_\zeta \zeta_{t-1} + \varepsilon_t^\zeta \quad (53)$$

$$u_t = \rho_u u_{t-1} + \varepsilon_t^u \quad (54)$$

Final Log-Linearized System:

$$\tilde{y}_t = E_t(\tilde{y}_{t+1}) - \frac{1}{\sigma}(\hat{i}_t - E_t(\hat{\pi}_{t+1})) + \gamma'_g(\tilde{g}_t - E_t \tilde{g}_{t+1}) + \zeta_t \quad (55)$$

$$\hat{\pi}_t = \beta(1 + dy)^{\frac{1}{\sigma}} E_t \hat{\pi}_{t+1} + \kappa \tilde{y}_t + u_t \quad (56)$$

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + (1 - \rho_i)(\phi_\pi^{\xi_t} \hat{\pi}_t + \phi_y^{\xi_t} \tilde{y}_t) + \varepsilon_t^{MP} \quad (57)$$

$$\tilde{g}_t = \rho_g \tilde{g}_{t-1} - \delta_b^{\xi_t} \hat{b}_t - \delta_g \tilde{y}_t + \varepsilon_t^g \quad (58)$$

$$\hat{b}_t = (1 + i_{ss} - \pi_{ss} - dy) \hat{b}_{t-1} \quad (59)$$

$$+ (i_{ss} + \hat{i}_{t-1} - \pi_{ss} - \hat{\pi}_t - dy - \tilde{y}_t + \tilde{y}_{t-1}) b_{ss} + g_{ss} + \tilde{g}_t \quad (60)$$

$$\zeta_t = \rho_\zeta \zeta_{t-1} + \varepsilon_t^\zeta \tag{61}$$

$$u_t = \rho_u u_{t-1} + \varepsilon_t^u \tag{62}$$

C Prior distribution of MS-DSGE

Table 8: Calibrated Parameters

Parameter	Description	Value
σ	Inverse intertemporal elasticity of substitution	1.00
β	Discount factor	0.99
κ	Slope of the Phillips curve	0.05
$\delta_b^{(2)}$	Fiscal feedback in regime 2	0.00
π_{ss}	Steady-state quarterly inflation	0.63%
dy	Deterministic quarterly growth rate	0.01

Table 9: Prior Distributions for Estimated Parameters

Parameter	Prior Mean	Initial Value	Std. Dev.	Distribution	Lower Bound	Upper Bound
ρ_r	0.75	0.75	0.05	Beta	0.30	0.98
ρ_u	0.50	0.50	0.10	Beta	0.00	0.95
ρ_g	0.60	0.60	0.10	Beta	0.00	0.98
ρ_ζ	0.80	0.80	0.08	Beta	0.30	0.99
b_{ss}	0.55	0.55	0.03	Normal	—	—
R_{ss}	0.010	0.010	0.002	Normal	—	—
g_{ss}	0.020	0.020	0.003	Normal	—	—
γ	0.08	0.08	0.03	Normal	0.00	0.50
δ_y	0.05	0.05	0.03	Normal	0.00	0.50
$\phi_\pi^{(1)}$	1.80	1.80	0.15	Normal	1.10	3.00
$\phi_\pi^{(2)}$	0.85	0.85	0.08	Normal	0.50	1.00
$\phi_y^{(1)}$	0.25	0.25	0.08	Normal	0.00	1.00
$\phi_y^{(2)}$	0.10	0.10	0.05	Normal	0.00	0.30
$\delta_b^{(1)}$	0.08	0.08	0.03	Gamma	0.01	0.50
σ_r	0.01	0.01	0.005	Inverse Gamma	—	—
σ_ζ	0.01	0.01	0.005	Inverse Gamma	—	—
σ_u	0.01	0.01	0.005	Inverse Gamma	—	—
σ_g	0.01	0.01	0.005	Inverse Gamma	—	—
σ_b	0.01	0.01	0.005	Inverse Gamma	—	—
$\sigma_{covid,y}$	0.03	0.03	0.01	Inverse Gamma	—	—
$\sigma_{covid,\pi}$	0.02	0.02	0.01	Inverse Gamma	—	—
p_{12}	0.02	0.02	0.01	Beta	—	—
p_{21}	0.12	0.12	0.03	Beta	—	—

D Results of MS-DSGE

The full posterior mean of VAR matrix is presented in table 10.

Table 10: Posterior Parameter Estimates						
Parameter	Mean	Median	LB68	UB68	LB90	UB90
ρ_r	0.8821	0.8832	0.8666	0.8976	0.8553	0.9063
ρ_u	0.3009	0.3001	0.2556	0.3471	0.2251	0.3749
ρ_g	0.5539	0.5543	0.4966	0.6128	0.4506	0.6483
ρ_ζ	0.9167	0.9183	0.9000	0.9337	0.8862	0.9428
b_{ss}	0.3032	0.3025	0.2626	0.3435	0.2403	0.3719
R_{ss}	0.0097	0.0097	0.0077	0.0117	0.0063	0.0131
g_{ss}	0.0034	0.0034	0.0021	0.0047	0.0013	0.0056
γ	0.0765	0.0765	0.0474	0.1052	0.0301	0.1241
δ_y	0.0840	0.0828	0.0583	0.1098	0.0451	0.1273
$\phi_{\pi,1}$	1.5761	1.5745	1.4061	1.7470	1.2975	1.8600
$\phi_{\pi,2}$	0.7130	0.7121	0.6312	0.7952	0.5838	0.8427
$\phi_{y,1}$	0.4099	0.4109	0.3598	0.4613	0.3226	0.4923
$\phi_{y,2}$	0.1440	0.1471	0.0892	0.1930	0.0562	0.2318
$\delta_{b,1}$	0.0480	0.0458	0.0319	0.0637	0.0248	0.0781
σ_r	0.0021	0.0021	0.0019	0.0023	0.0018	0.0024
σ_ζ	0.0030	0.0029	0.0025	0.0035	0.0023	0.0039
σ_u	0.0046	0.0046	0.0043	0.0049	0.0041	0.0052
σ_g	0.0083	0.0083	0.0077	0.0090	0.0073	0.0095
σ_b	0.0140	0.0140	0.0131	0.0151	0.0125	0.0158
$\sigma_{covid,y}$	0.0408	0.0394	0.0275	0.0546	0.0222	0.0645
$\sigma_{covid,\pi}$	0.0172	0.0165	0.0118	0.0226	0.0095	0.0272
$snk_{tp,1,2}$	0.0323	0.0305	0.0201	0.0445	0.0147	0.0566
$snk_{tp,2,1}$	0.1309	0.1292	0.1014	0.1605	0.0864	0.1815